

reflection

- A *reflection* is when a wave abruptly changes direction.
- A wave is reflected whenever it reaches a boundary of its medium or encounters an abrupt change in the properties of its medium.

References:

Physics, Johnson & Cutnell

Physics for future presidents, Richard Muller



Reflection of light (visible or invisible)

Angle of incidence = angle of reflection.

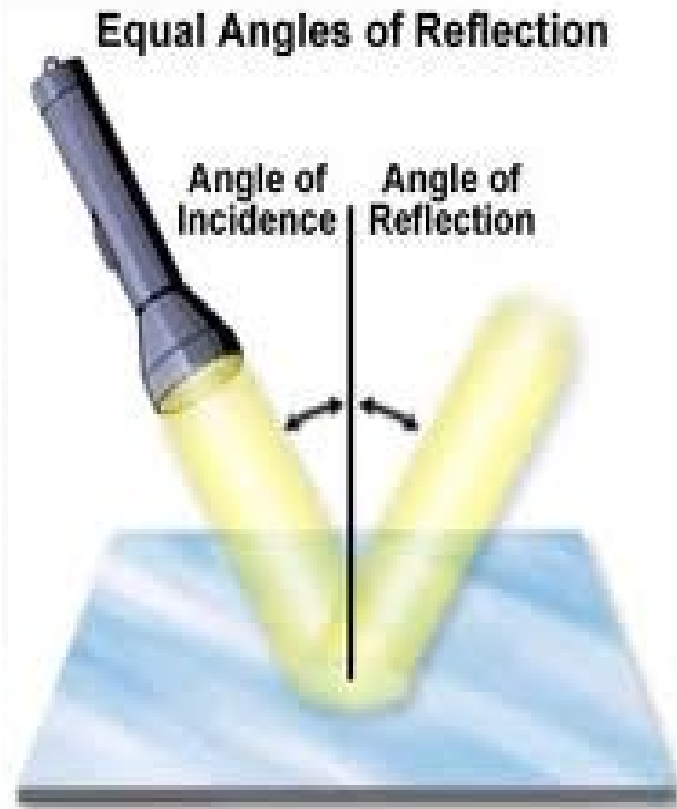
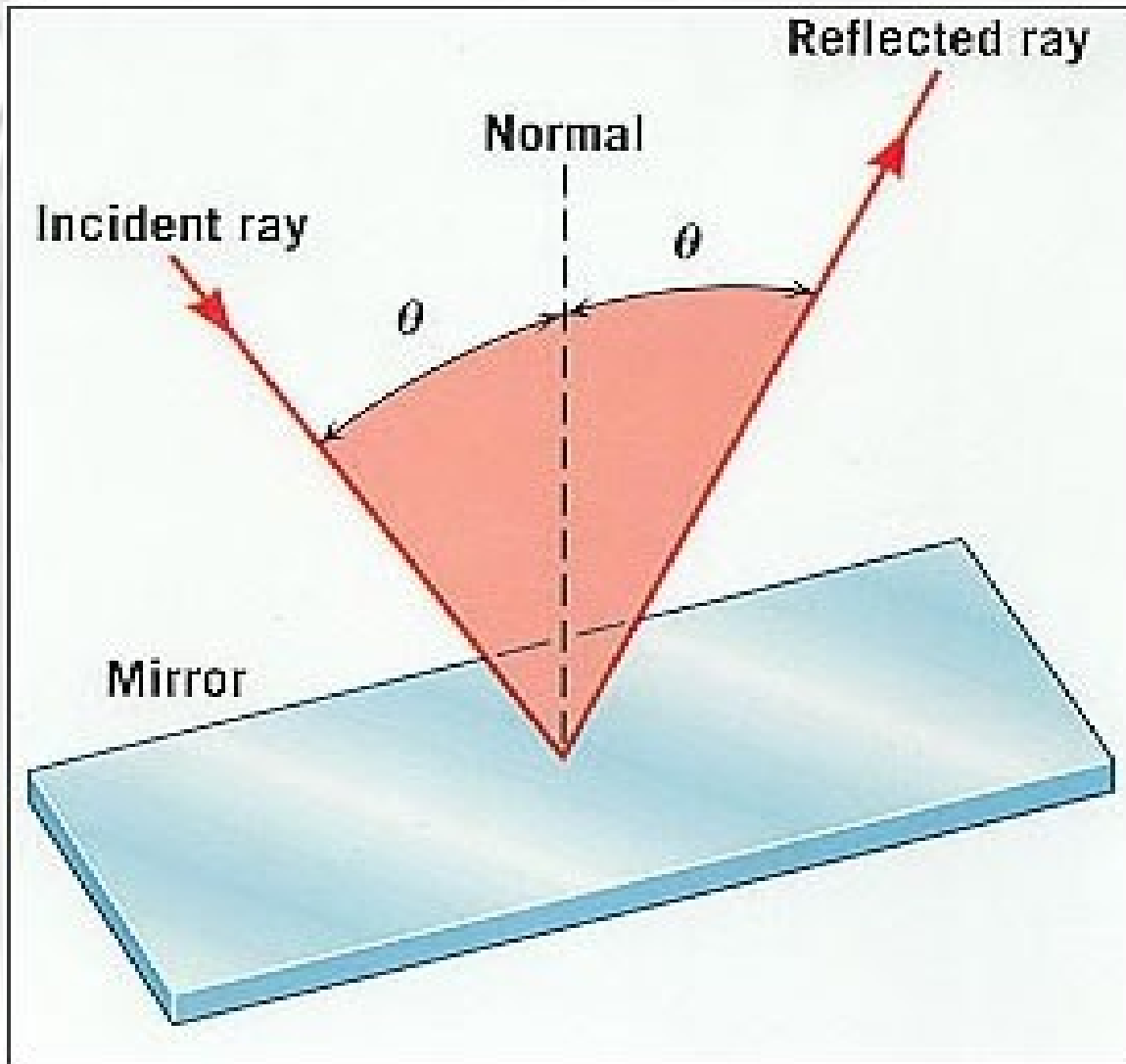
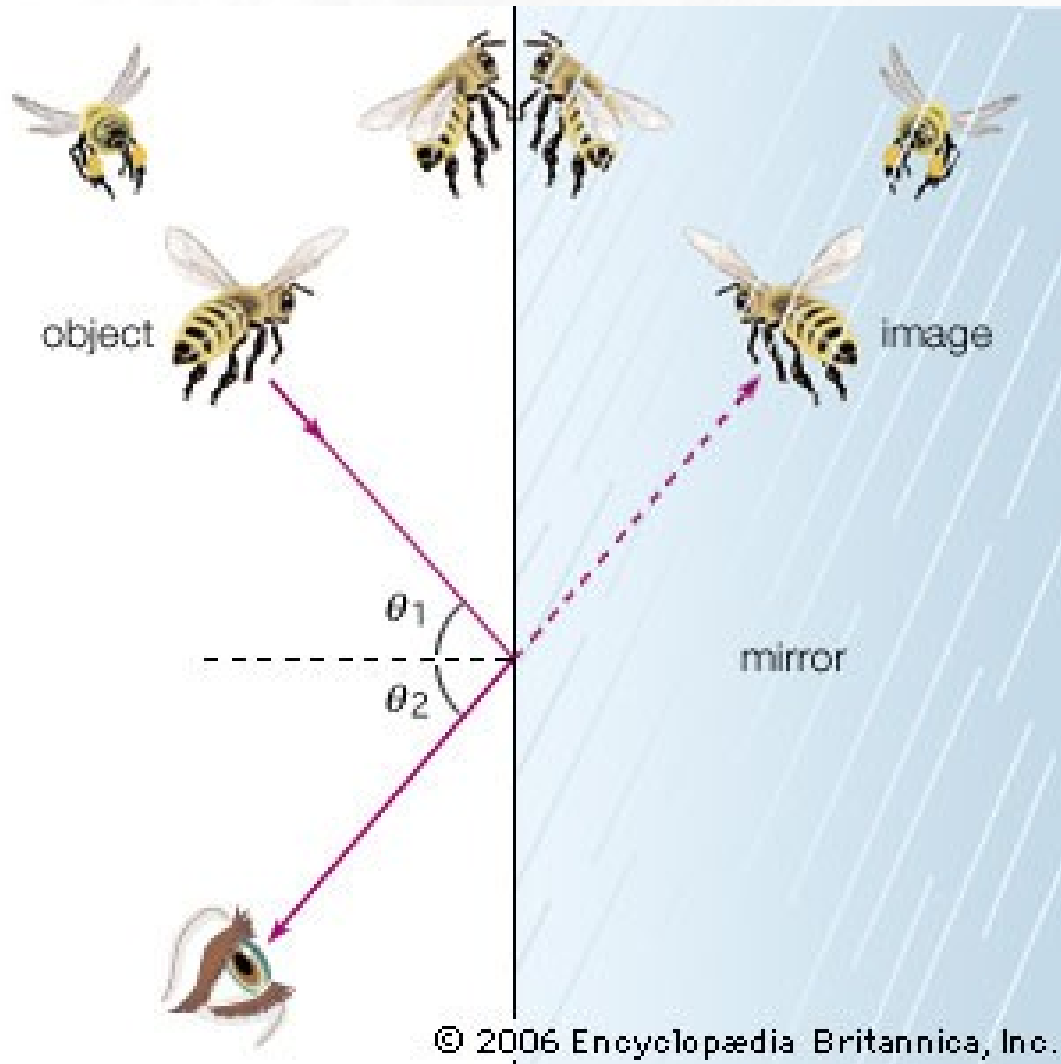
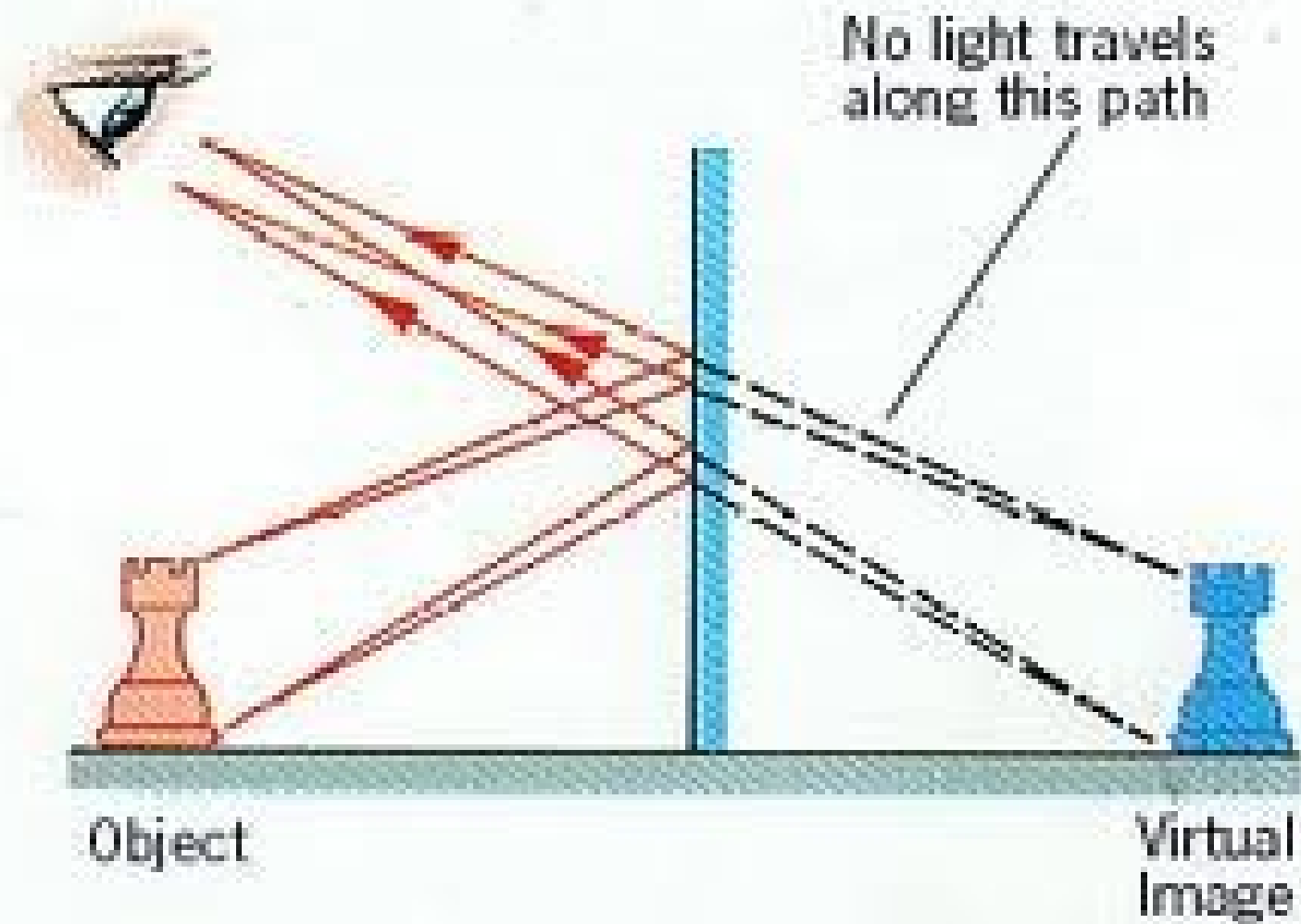


Figure 2

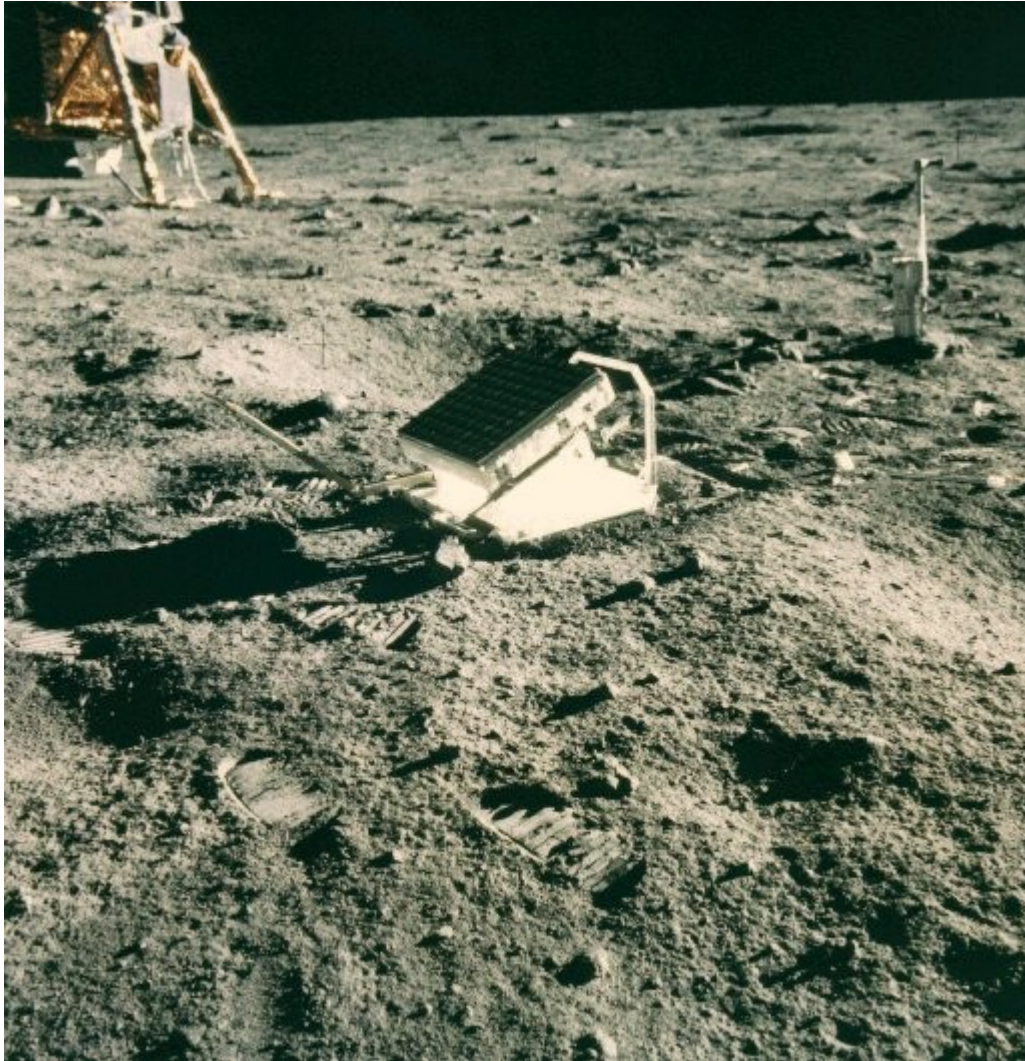
Reflection of light (visible or invisible)

Angle of incidence = angle of reflection.



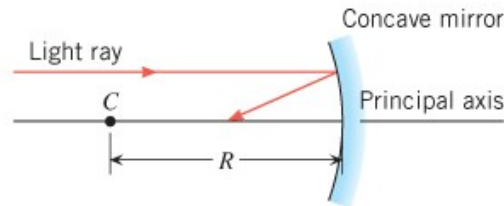
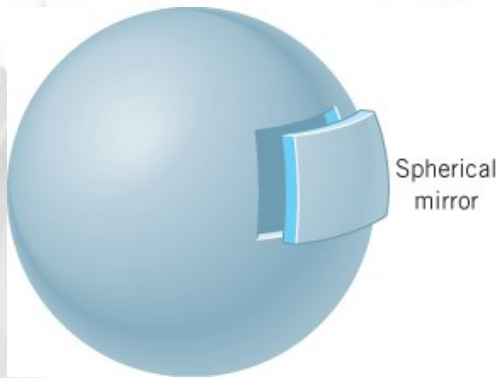


The top of a mirror is placed at your eye level. If you want to see your feet in the mirror, how long the mirror must be?
If the mirror is shorter, why can't you do to see the feet? draw.

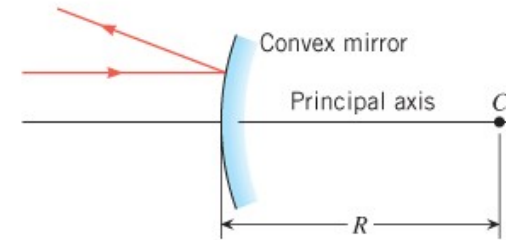


Retro reflector on the Moon.
A laser ray from Earth
Will bounce back.
We can compute the distance
Earth-Moon with precision.
384,400 km in average.

Reflection off a curved mirror:



$$f = \frac{1}{2} R$$



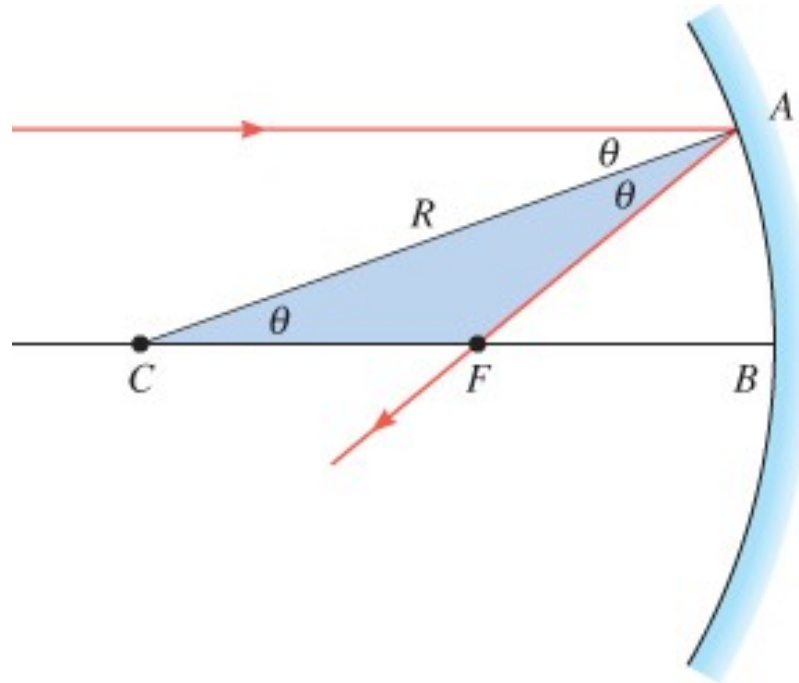
$$f = -\frac{1}{2} R$$

If the inside surface of the spherical mirror is polished, it is a **concave mirror**. If the outside surface is polished, is it a **convex mirror**.

The law of reflection applies, just as it does for a plane mirror.

The **principal axis** of the mirror is a straight line drawn through the center and the midpoint of the mirror.

25.4 Spherical Mirrors

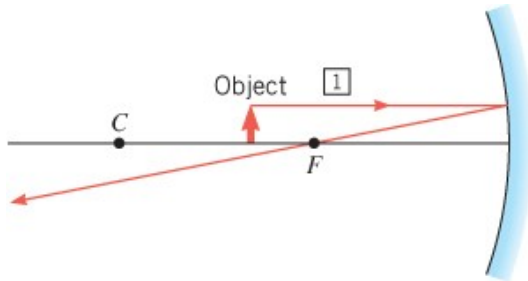


The focal point of a concave mirror is halfway between the center of curvature of the mirror C and the mirror at B .

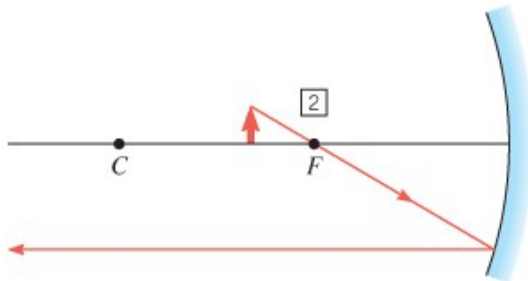
$$f = \frac{1}{2} R$$

25.5 The Formation of Images by Spherical Mirrors

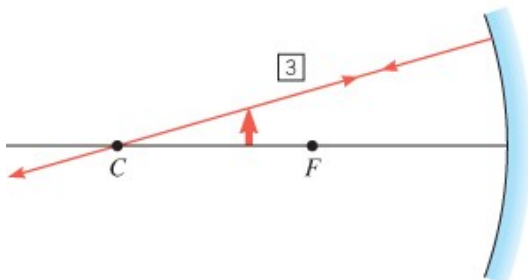
CONCAVE MIRRORS



This ray is initially parallel to the principal axis and passes through the focal point.



This ray initially passes through the focal point, then emerges parallel to the principal axis.



This ray travels along a line that passes through the center.

Image formation and the principle of reversibility

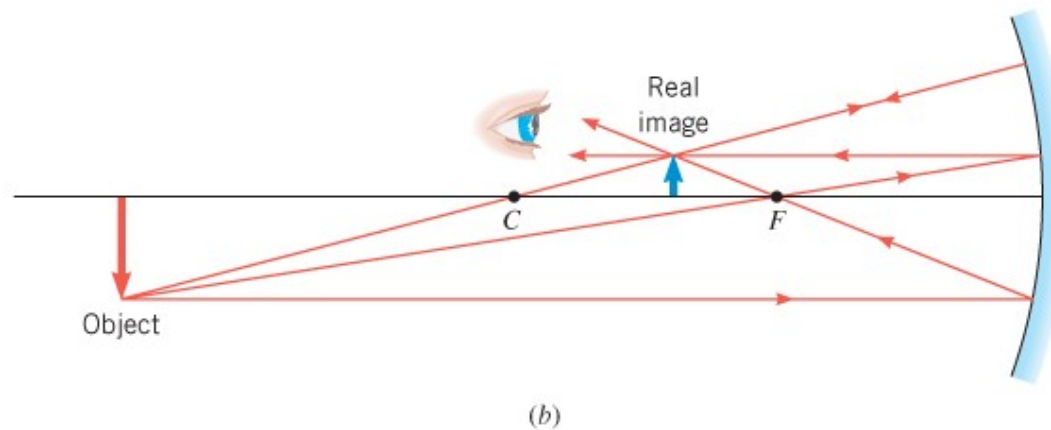
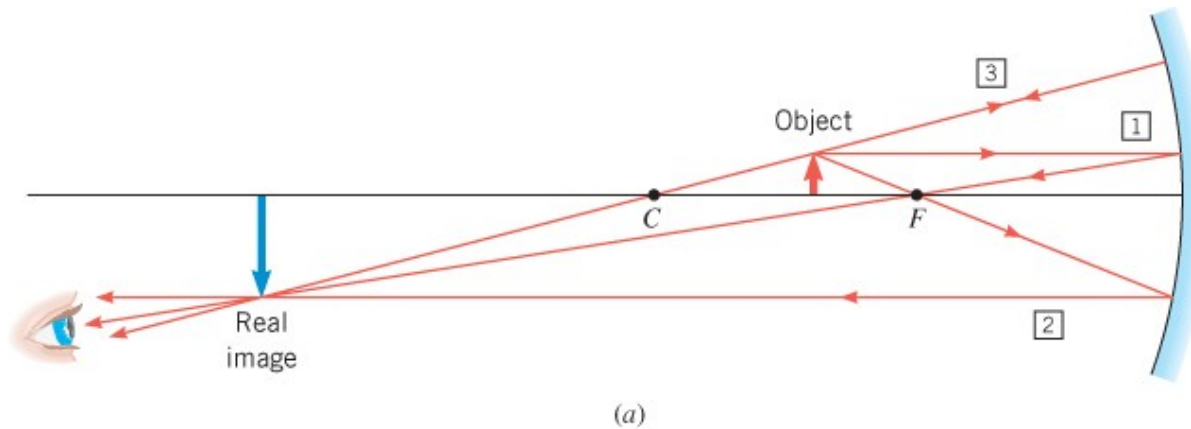
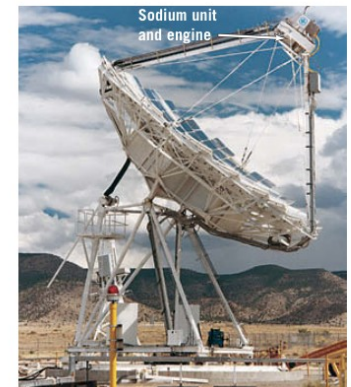


Figure 25.15 This long row of parabolic mirrors focuses the sun's rays to heat an oil-filled pipe located at the focal point of each mirror. Many such rows are used by a solar-thermal electric plant in the Mojave Desert. (© Jim West/Alamy)



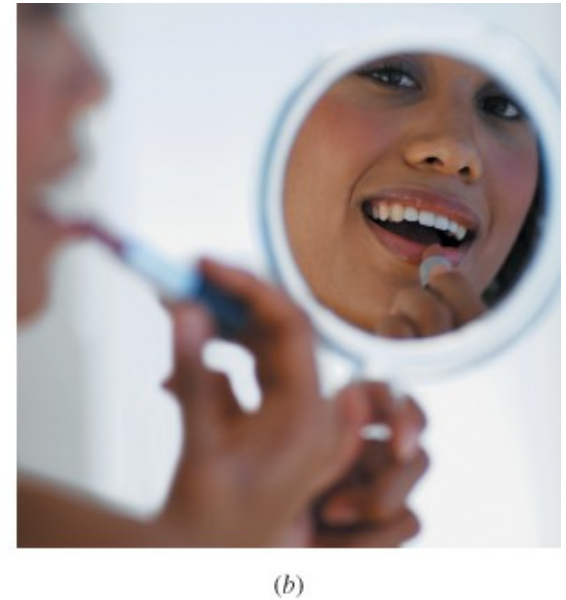
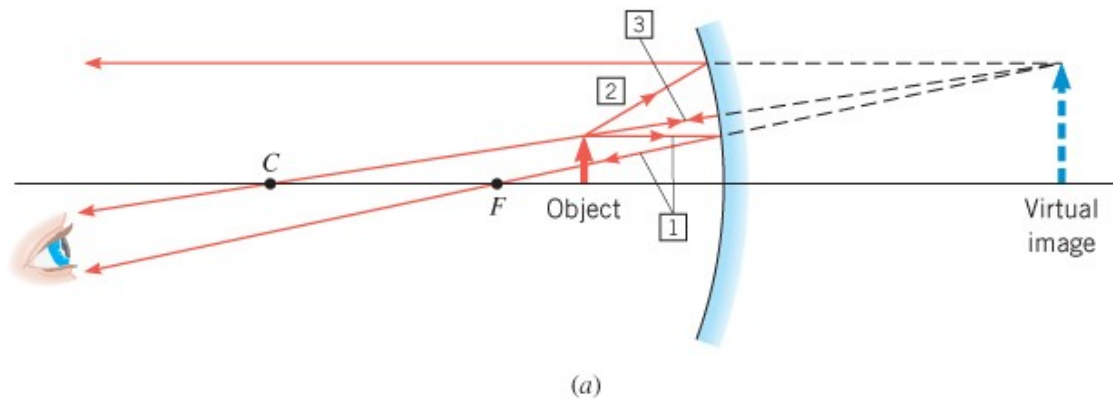
5. The photograph shows an experimental device at Sandia National Laboratories in New Mexico. This device is a mirror that focuses sunlight to heat sodium to a boil, which then heats helium gas in an engine. The engine does the work of driving a generator to produce electricity. The sodium unit and the engine are labeled in the photo. (a) What kind of mirror, concave or convex, is being used? (b) Where is the sodium unit located relative to the mirror? Express your answer in terms of the focal length of the mirror.
6. Refer to Figure 25.14 and the related discussion about spherical aberration. To bring the top ray closer to the focal point F after reflection, describe how you would change the shape of the mirror. Would you open it up to produce a more gently curving shape or bring the top and bottom edges closer to the principal axis?



(Courtesy Sandia National Laboratories)

Question 5

25.5 The Formation of Images by Spherical Mirrors

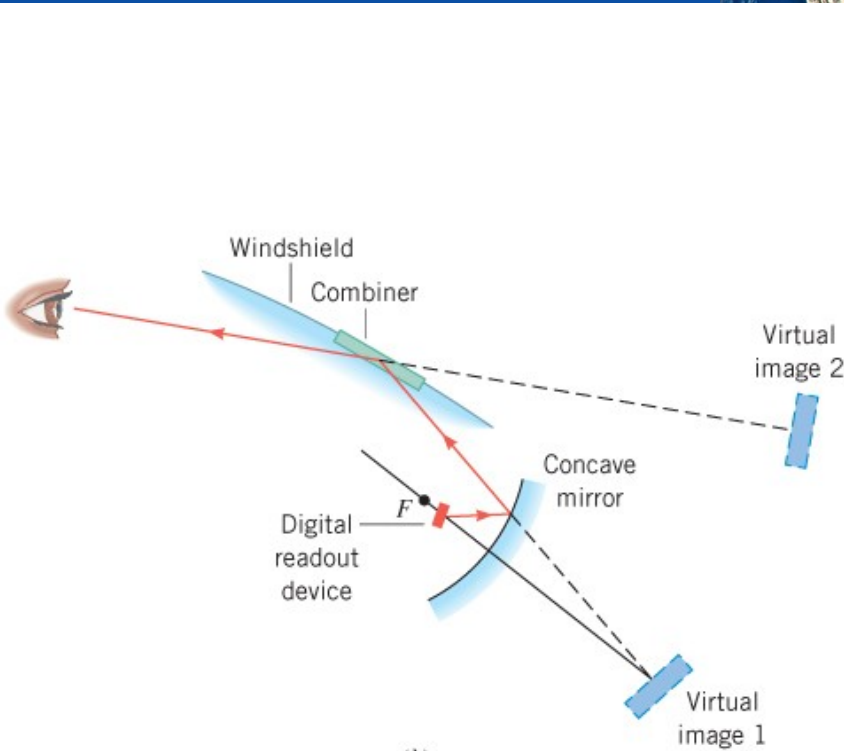


When an object is located between the focal point and a concave mirror, an enlarged, upright, and virtual image is produced.

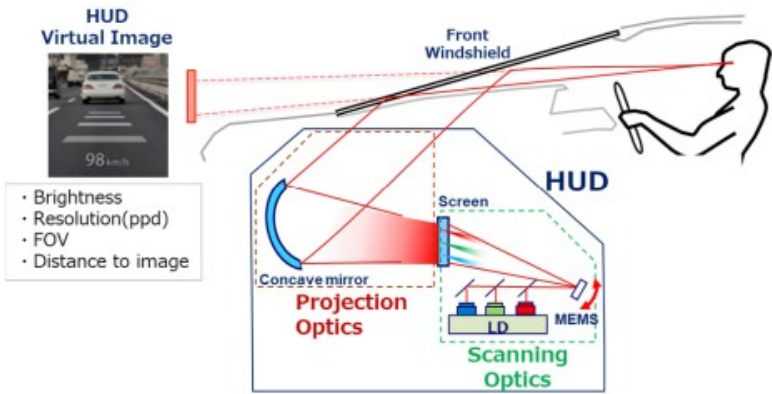
25.5 The Formation of Images by Spherical Mirrors



(a)



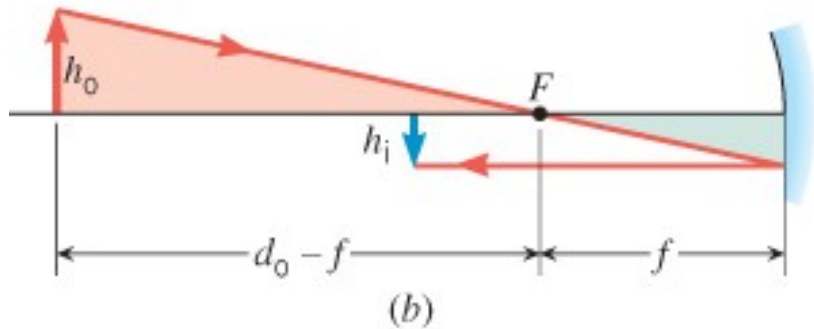
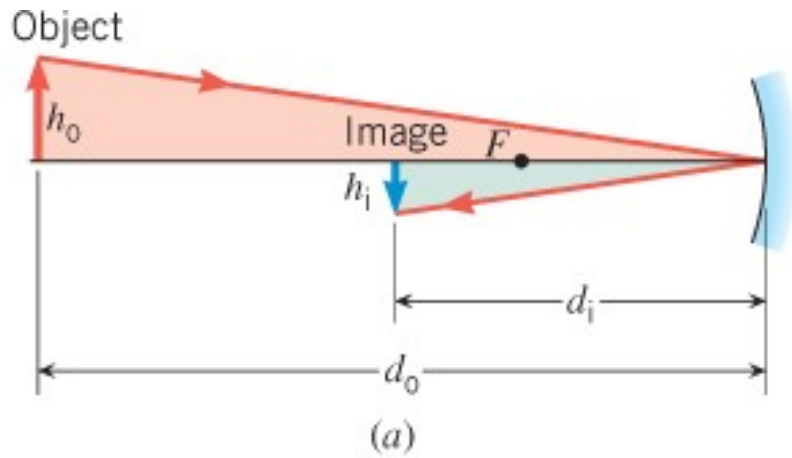
(b)



- Brightness
• Resolution(ppd)
• FOV
• Distance to image

25.6 The Mirror Equation and Magnification

These diagrams are used to derive the mirror equation.



$$\frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}$$

$$m = \frac{h_i}{h_o} = -\frac{d_i}{d_o}$$

Example 3 A Real Image Formed by a Concave Mirror

A 2.0-cm-high object is placed 7.10 cm from a concave mirror whose radius of curvature is 10.20 cm. Find (a) the location of the image and (b) its size.

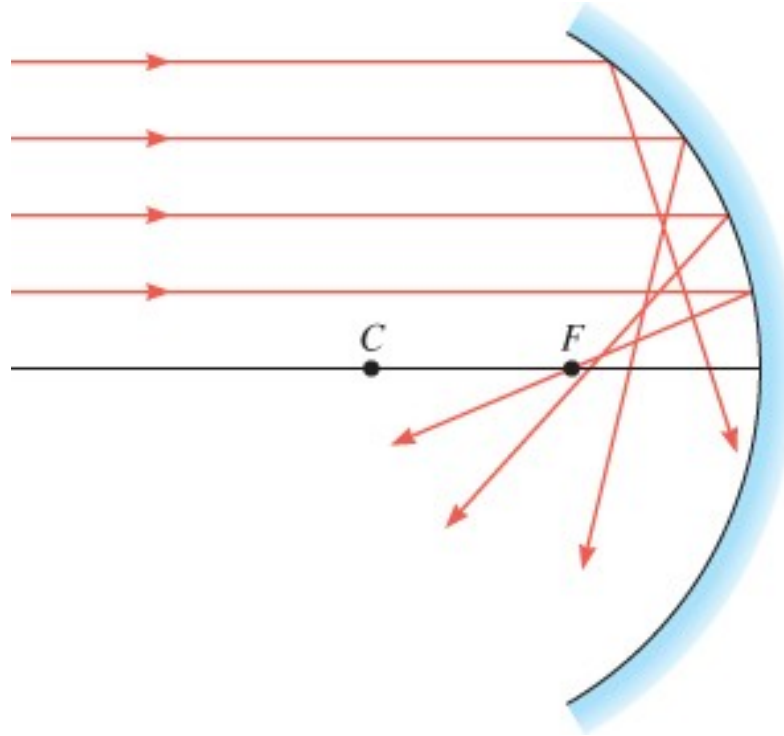
Reasoning For a concave mirror, Equation 25.1 gives the focal length as $f = \frac{1}{2}R$. Therefore, the focal length is $f = \frac{1}{2}(10.20 \text{ cm}) = 5.10 \text{ cm}$, and the object is located between the focal point F and the center of curvature C of the mirror, as in Figure 25.18a. Based on this figure, we expect that the image is real and that, relative to the object, it is farther away from the mirror, inverted, and larger.

Example 4 A Virtual Image Formed by a Concave Mirror

An object is placed 6.00 cm in front of a concave mirror that has a 10.0-cm focal length. (a) Determine the location of the image. (b) The object is 1.2 cm high. Find the image height.

Reasoning The object is located between the focal point and the mirror, as in Figure 25.19a. The setup is analogous to a person using a makeup or shaving mirror. Therefore, we expect that the image is virtual and that, relative to the object, it is upright and larger.

25.4 Spherical Mirrors

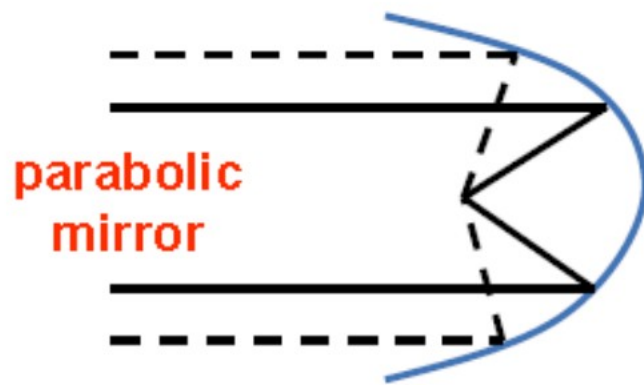
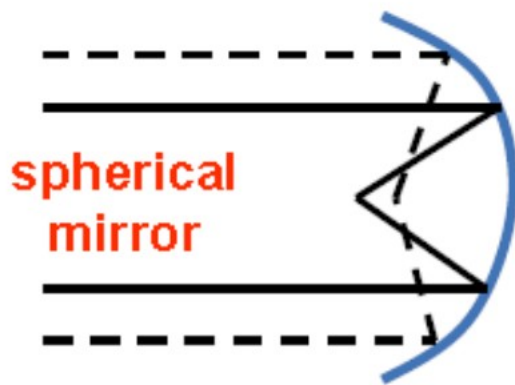


Rays that lie close to the principal axis are called paraxial rays.

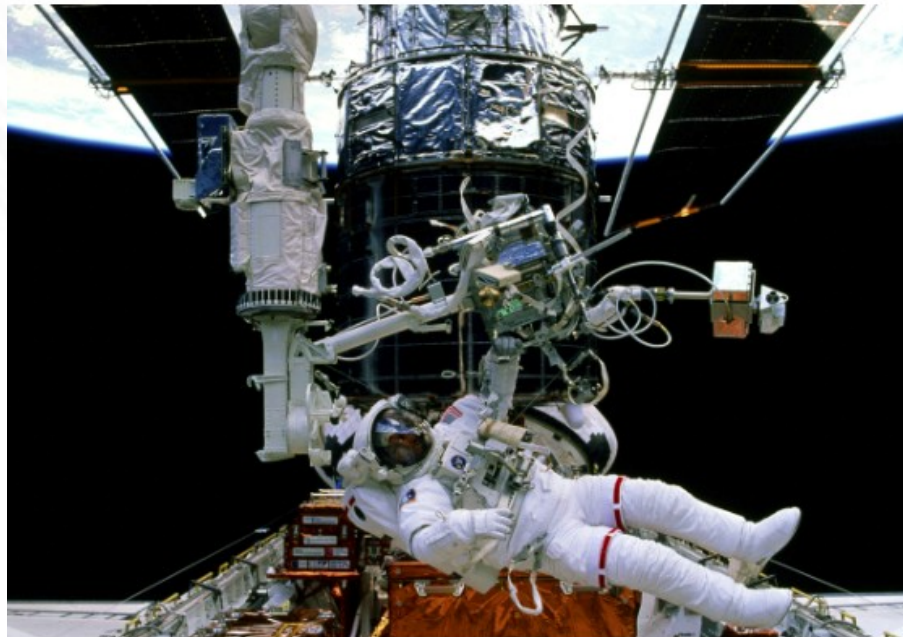
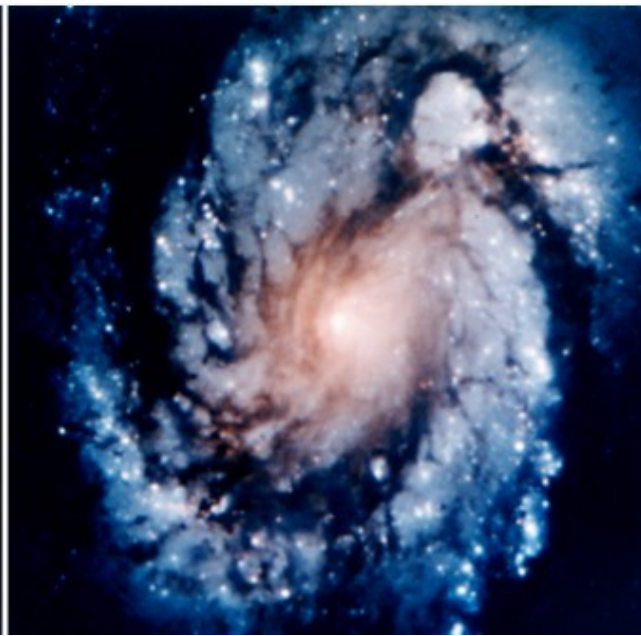
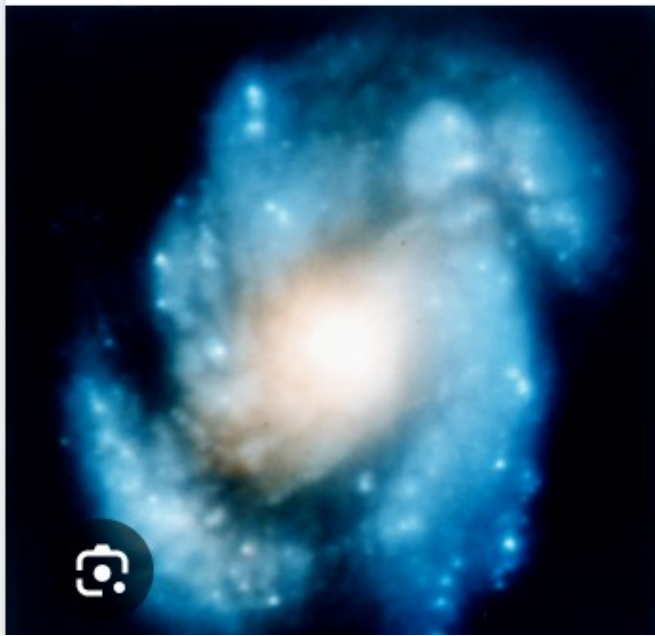
Rays that are far from the principal axis do not converge to a single point. The fact that a spherical mirror does not bring all parallel rays to a single point is known as ***spherical aberration***.

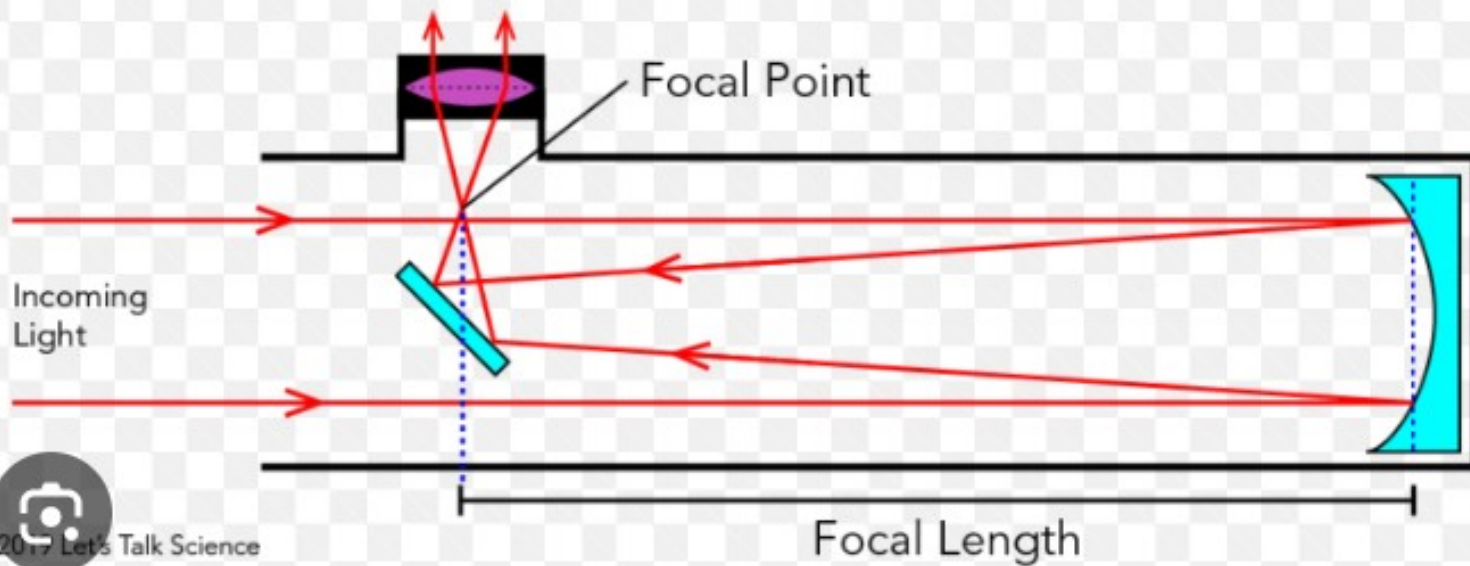
Reflecting Telescopes

- Use a **mirror** to bring light to a focus.



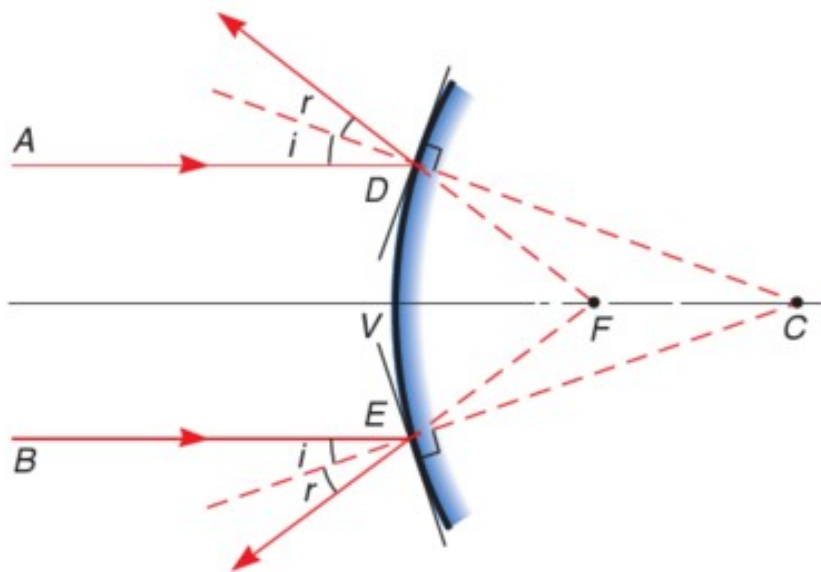
- A spherical mirror does not focus the light to a single point, which leads to **spherical aberration**.
- A parabolic mirror does focus the light to a single point.
- This was in fact the problem that Hubble first had.





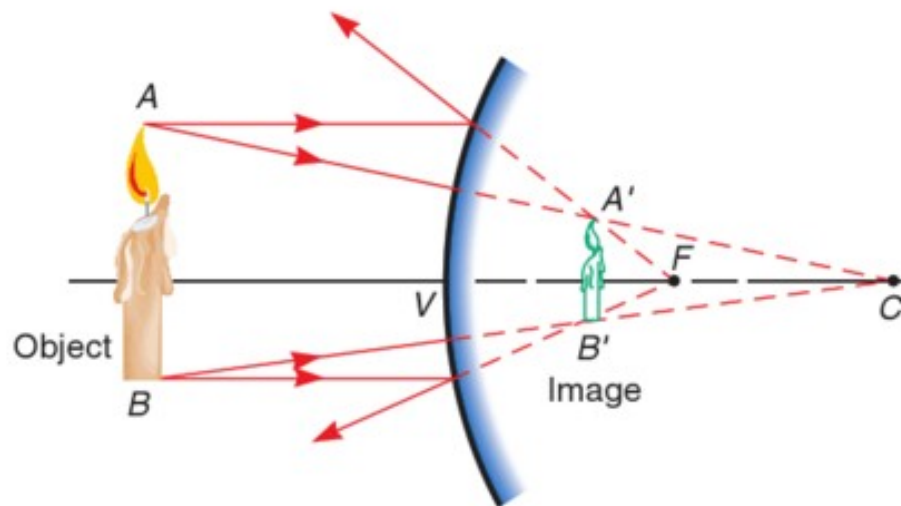
The first reflective Telescope was made By Isaac Newton and Donated to the Royal Society of London. 1660. this is How Newton Became a member of The society.
(replica by F L Agate 1924)

Figure 21.15 Formation of images in convex mirrors.



Note that $\angle i = \angle r$ for convex mirrors also.

(a)

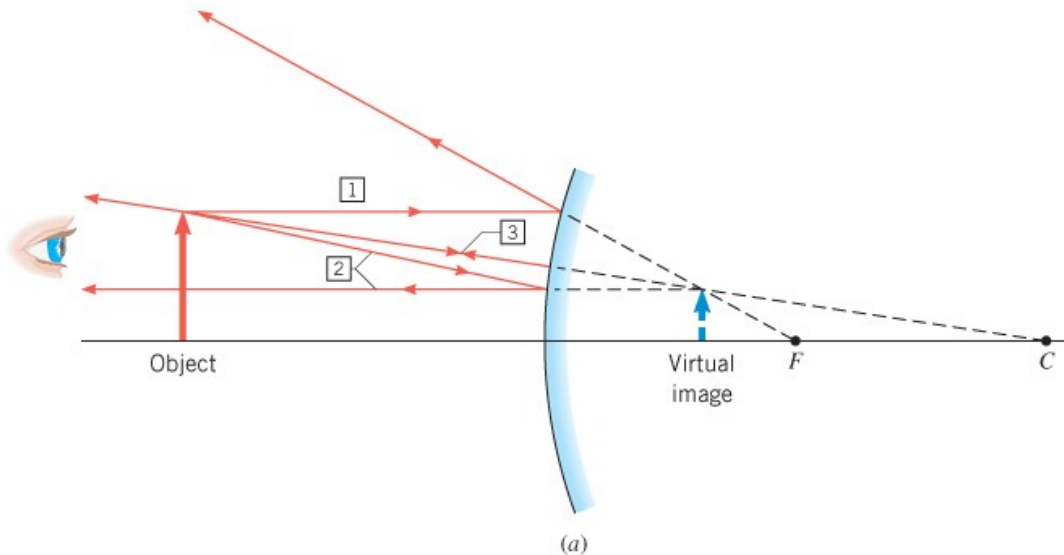


$$f = -\frac{1}{2} R$$



25.5 The Formation of Images by Spherical Mirrors

CONVEX MIRRORS



Ray 1 is initially parallel to the principal axis and appears to originate from the focal point.

Ray 2 heads towards the focal point, emerging parallel to the principal axis.

Ray 3 travels toward the center of curvature and reflects back on itself.

25.6 *The Mirror Equation and Magnification*

Summary of Sign Conventions for Spherical Mirrors

f is + for a concave mirror.

f is – for a convex mirror.

d_o is + if the object is in front of the mirror.

d_o is – if the object is behind the mirror.

d_i is + if the object is in front of the mirror (real image).

d_i is – if the object is behind the mirror (virtual image).

m is + for an upright object.

m is – for an inverted object.

25.6 The Mirror Equation and Magnification

Example 5 A Virtual Image Formed by a Convex Mirror

A convex mirror is used to reflect light from an object placed 66 cm in front of the mirror. The focal length of the mirror is -46 cm. Find the location of the image and the magnification.

... if the mirror were planar, as Example 5 illustrates.

Example 6 A Convex Versus a Plane Mirror

An object is placed 9.00 cm in front of a mirror. The image is 3.00 cm closer to the mirror when the mirror is convex than when it is planar (see Figure 25.23). Find the focal length of the convex mirror.

Reasoning For a plane mirror, the image and the object are the same distance on either side of the mirror. Thus, the image would be 9.00 cm behind a plane mirror. If the image in a convex mirror is 3.00 cm closer than this, the image must be located 6.00 cm behind the convex mirror. In other words, when the object distance is $d_o = 9.00$ cm, the image distance for the convex mirror is $d_i = -6.00$ cm (negative because the image is virtual). The mirror equation can be used to find the focal length of the mirror.

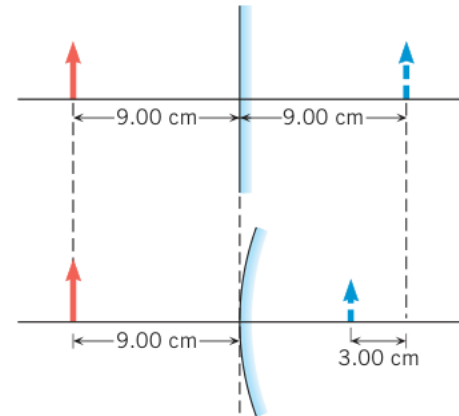


Figure 25.23 The object distance (9.00 cm) is the same for the plane mirror (top part of

Example 5 A Virtual Image Formed by a Convex Mirror

A convex mirror is used to reflect light from an object placed 66 cm in front of the mirror. The focal length of the mirror is -46 cm. Find the location of the image and the magnification.

$$\frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o} = \frac{1}{-46 \text{ cm}} - \frac{1}{66 \text{ cm}} = -0.037 \text{ cm}^{-1}$$

$$d_i = -27 \text{ cm}$$

$$m = -\frac{d_i}{d_o} = -\frac{(-27 \text{ cm})}{66 \text{ cm}} = 0.41$$

Example 7



The Physics of Keratometers

A contact lens rests against the cornea of the eye. Figure 25.24 shows an optometrist using a keratometer to measure the radius of curvature of the cornea, thereby ensuring that the prescribed lenses fit accurately. In the keratometer, light from an illuminated object reflects from the corneal surface, which acts like a convex mirror and forms an upright virtual image that is smaller than the object (see Figure 25.21*a*). With the object placed 9.0 cm in front of the cornea, the magnification of the corneal surface is measured to be 0.046. Determine the radius of the cornea.

Reasoning The radius of a convex mirror can be determined from the mirror's focal length, since the two are related. The focal length is related to the distances of the object and its image from the mirror via the mirror equation. The magnification of the mirror is also related to the object and image distances according to the magnification equation. By using the mirror equation and the magnification equation, we will be able to determine the focal length and, hence, the radius.

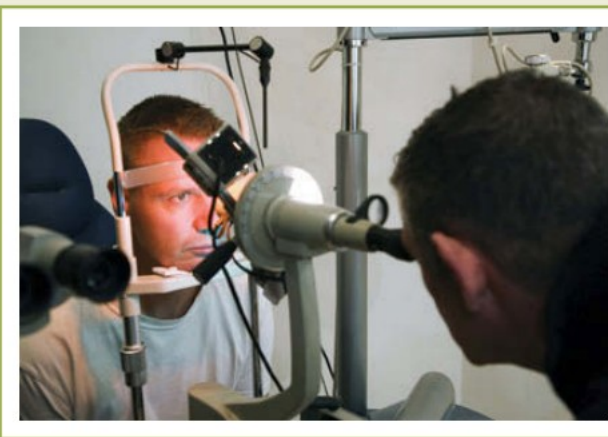


Figure 25.24 An optometrist is using a keratometer to measure the radius of curvature of the cornea of the eye, which is the surface against which a contact lens rests. (© Mark Thomas/SPL/Photo Researchers)

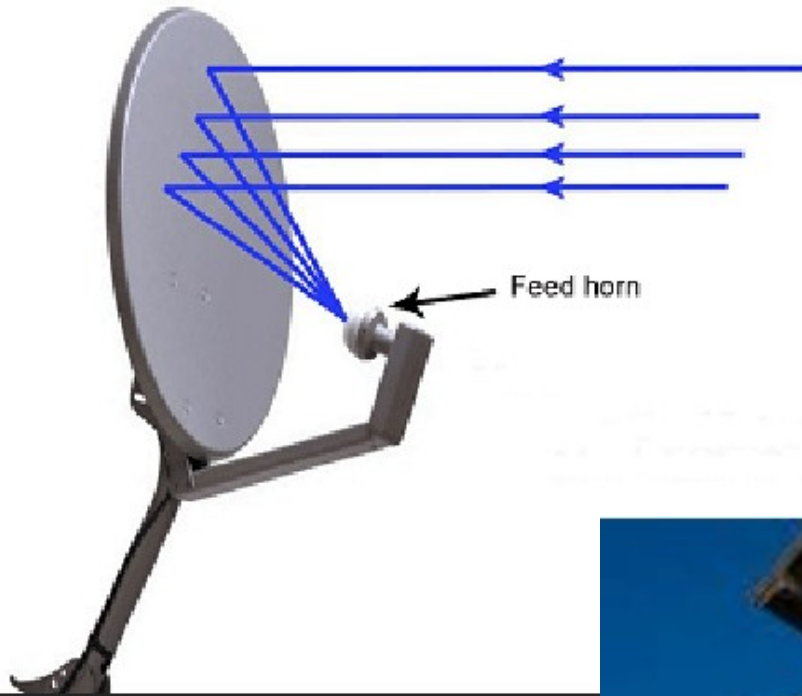
Two Choices

The radius of curvature of a mirror is 24 cm. A diamond ring is placed in front of this mirror. The image is twice the size of the ring. Find the object distance of the ring.

Finding the Focal Length

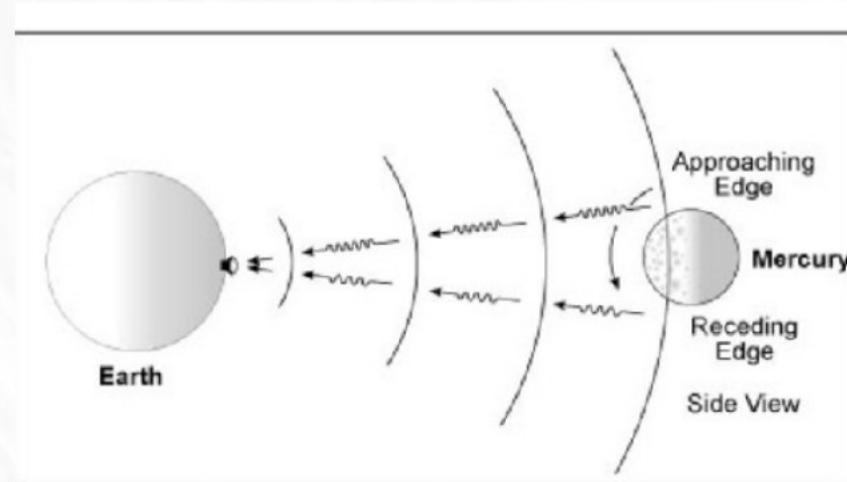
An object is located 7.0 cm in front of a mirror. The virtual image is located 4.5 cm away from the mirror and is smaller than the object. Find the focal length of the mirror.

Satellite dish/ Reflector



By measuring the time taken for the radar echo to come back, the distance can be calculated, since radio waves travel at the speed of light. Once this Earth-Venus distance is known, the distance between Earth and the Sun can be calculated

<https://history.nasa.gov/SP-4218/ch2.htm>



[27] In 1958, MIT's Lincoln Laboratory announced that it had bounced radar waves off Venus. That apparent success was followed by another, but in England, during Venus' next inferior conjunction. In September 1959, investigators at Jodrell Bank announced that they had validated the 1958 results, yet Lincoln Laboratory failed to duplicate them. All uncertainty was swept aside, when the Jet Propulsion Laboratory (JPL) obtained the first unambiguous detection of echoes from Venus in 1961

For example, astronomers use radio telescope to collect the Radio waves from space. The radio waves are focused on 1 point (the focal point) and collected. In the US there is The “ very large array” of telescopes in Arizona. (see the movie contact)



The idea is many of these Telescopes is equivalent To a huge one. We need huge dishes to compensate for The large wavelength of radio Waves. Large wavelength Means small resolution. The resolution of a telescope Is proportional to the diameter Of the objective (here dish) but Inversely proportional to the wavelength.

The waves Collected by the individual dishes are processed by a computer ²⁷
And every thing happen like it was collected by 1 huge dish.



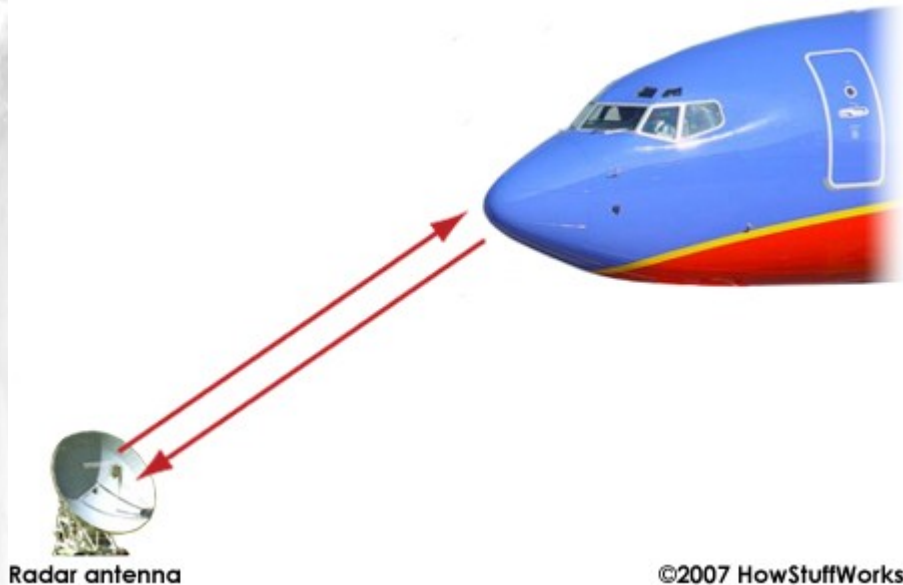
ARECIBO RADIO TELESCOPE (NAIC / NSF)

Here is a huge one (305m in diameter) in Puerto Rico. It is suspended in a valley. Used in the movie James Bobs, golden eye.

<http://www.youtube.com/watch?v=Mh2qnn4D0kY>

Reflection of light.

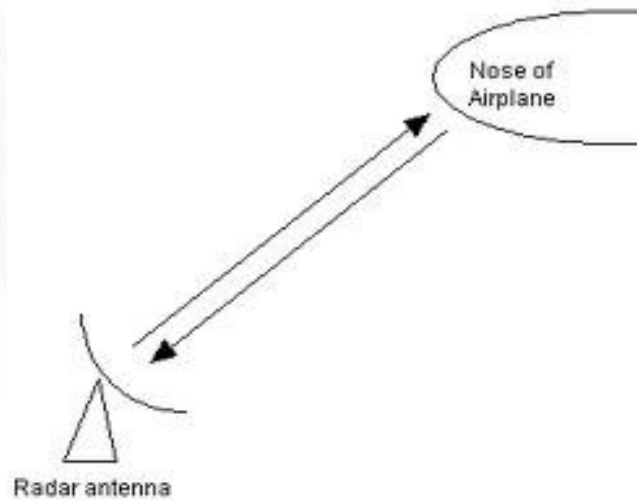
Radars use the refrection of radio waves to detect objects.



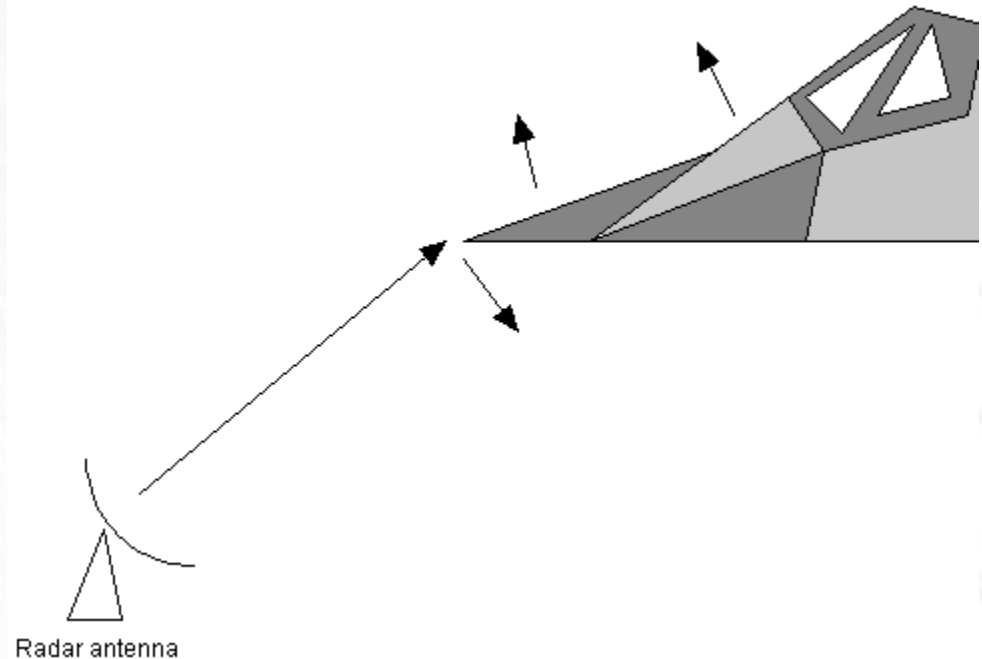
Radars were developed during WWII in England and was a military Secret. British pilots could detect the Presence of a German plane And attack it. Nazis had no idea What was going one. They Thought British planes were In great number. They gave up on Conquering England.

Now the army develop planes “invisible” to radars/ The radio waves are Reflected but in all direction but the one along which the detector is. This is called the **stealth technology**.

stealth technology

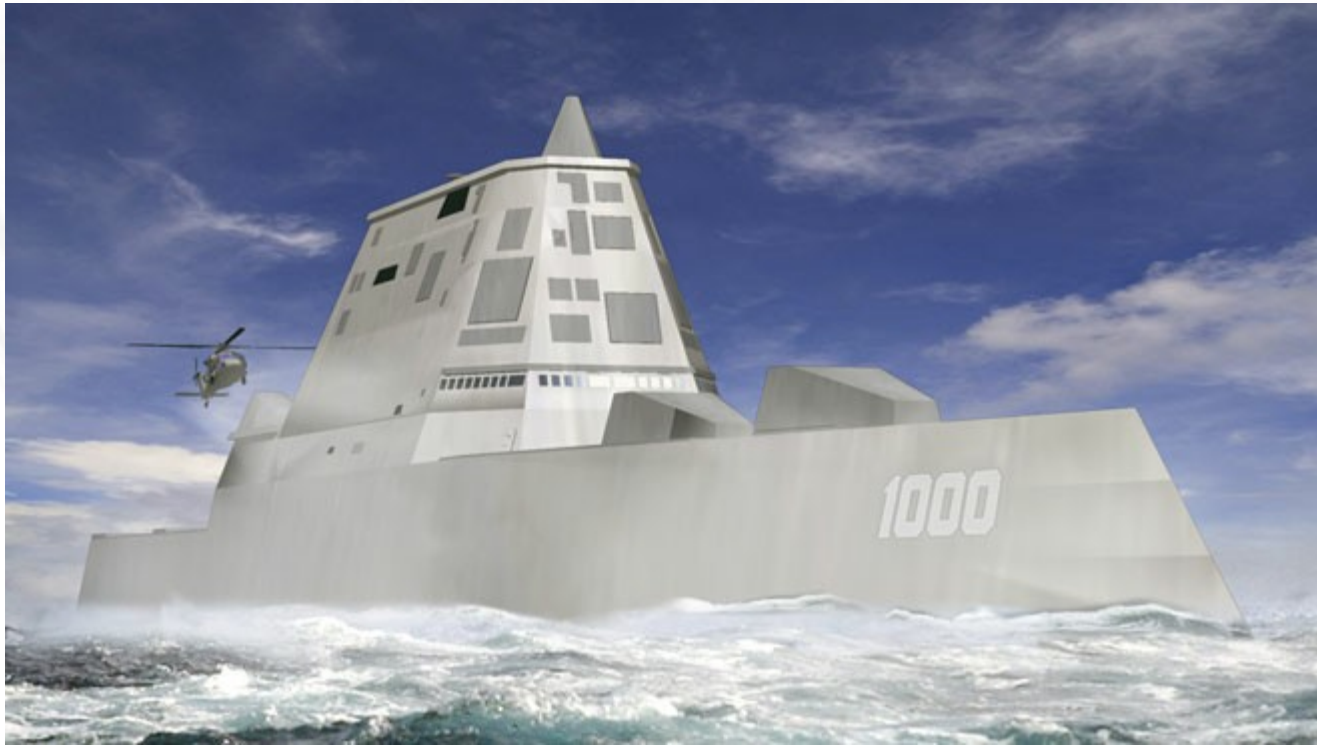


Smooth surface, you detect
The plane



The waves does not bounce back
In the same direction

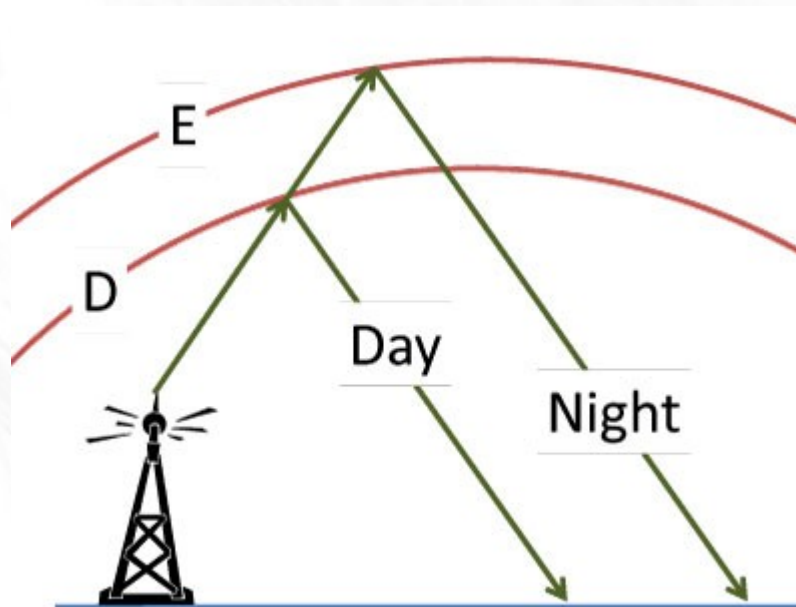




This file image released by Bath Iron Works shows a rendering of the DDG-1000 Zumwalt, the U.S. Navy's next-generation destroyer, which has been funded to be built at Bath Iron Works in Maine and at Northrop Grumman's shipyard in Pascagoula, Miss. (AP/Bath Iron Works)

Read more: <http://www.foxnews.com/politics/2012/06/04/planned-stealth-destroyer-could-underpin-us-navy-china-strategy/#ixzz1wpVlrtgY>

Radio waves are part of the EM spectrum. So it's light.
Radio waves are “invisible light” and reflect off boundaries too.
Below: AM radio waves reflect of layers of the atmosphere.



Water and bleach in bottle of coke was fitted on the roof. It reflects and refracts sunlight and is equal to 55 watts.

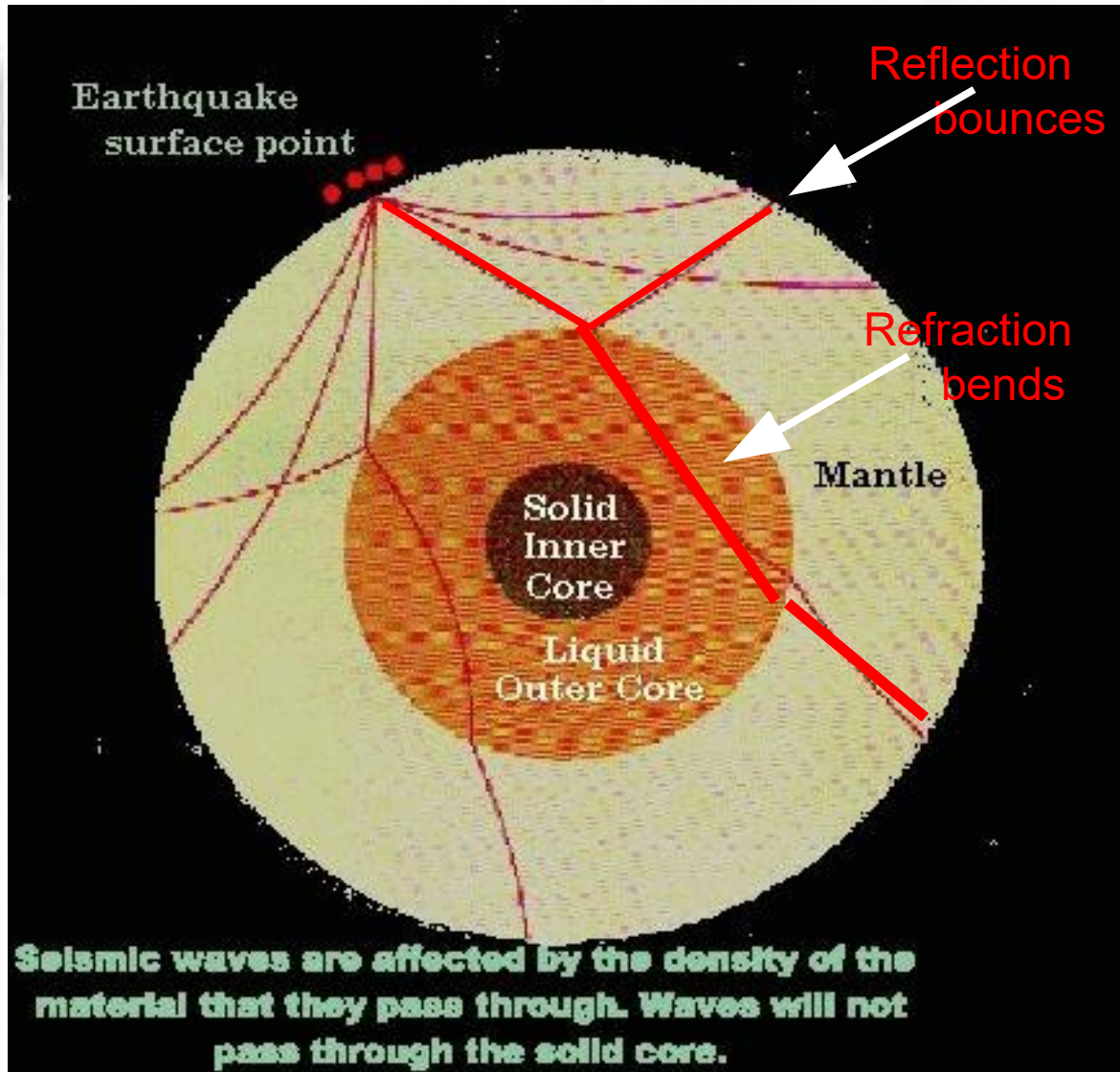


REFLECTION and REFRACTION (bending of waves)

When waves reach a different medium, they can bend (curve) or they can be reflected

http://phys23p.sl.psu.edu/phys_anim/waves/s_p_waves.avi s_p_waves

To review seismic waves. Longitudinal waves (P waves) and transverse (S waves)



Thanks to the Properties of Reflection and Bending we Can probe the Interior of the Earth.

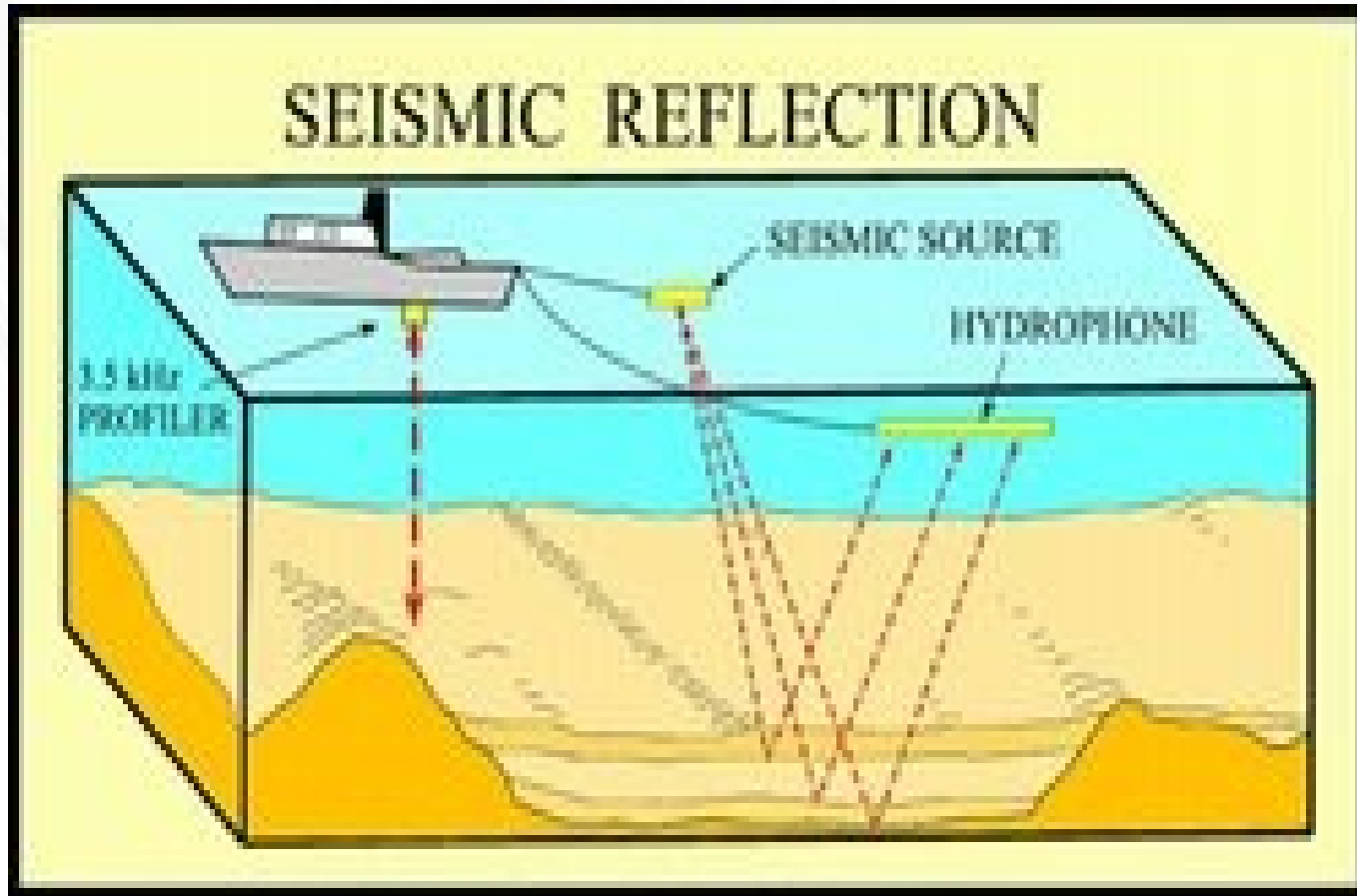
We infer the structure Of the Earth.

Oil company uses these Properties to find Oil reservoirs.

Which of the waves Are bending (curving)?
Which ones are Being reflected ?

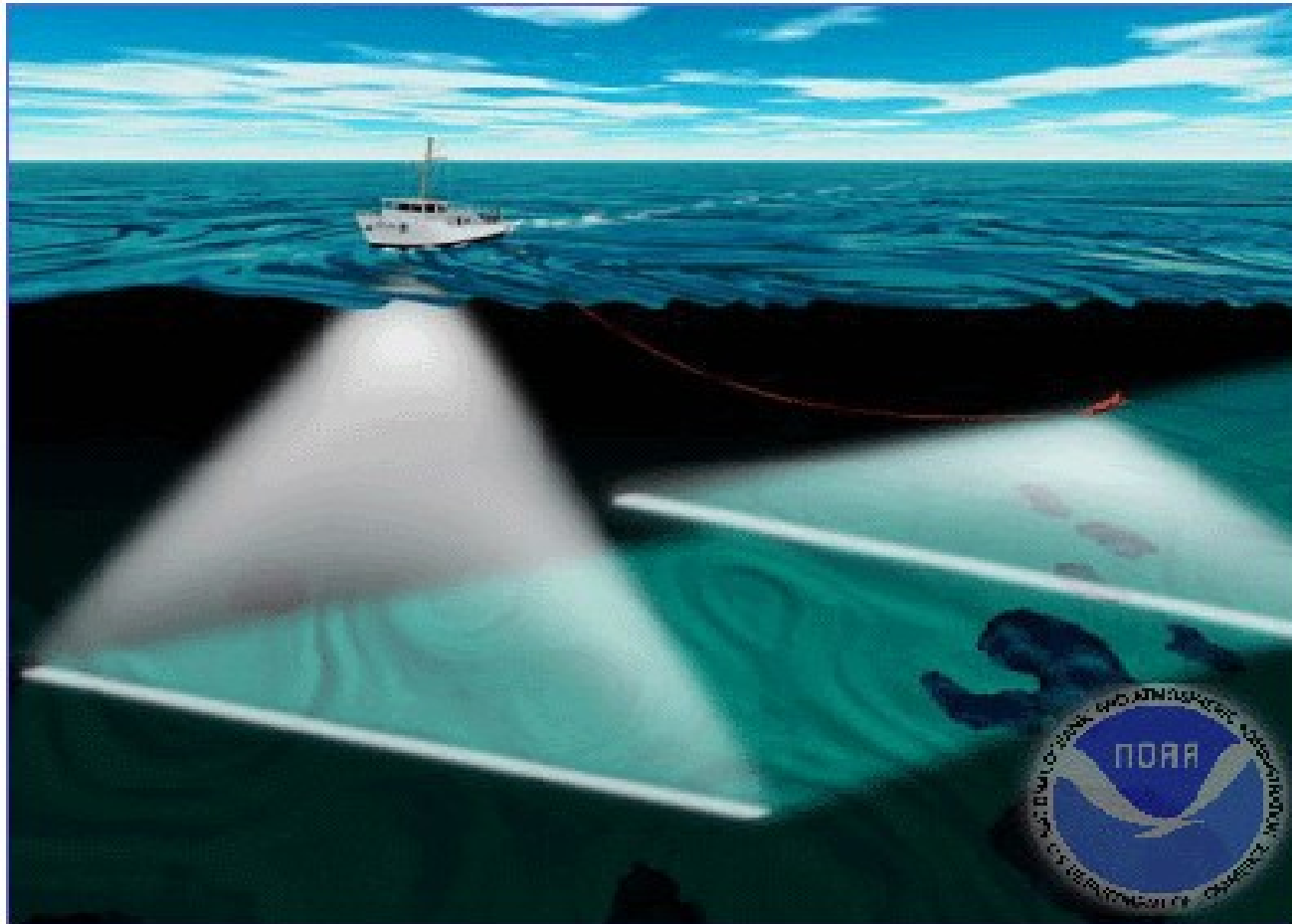
**From sound waves to seismic waves (P waves).
Reflection of waves are used to probe the ground.
P waves and sound waves are longitudinal waves.**

angle of incidence=angle of reflection



**BACK to sound waves. Sound waves can be reflected off
A boundary between 2 medium. If the direction of propagation is
Perpendicular to the surface, angle of incidence=angle of reflection =0**



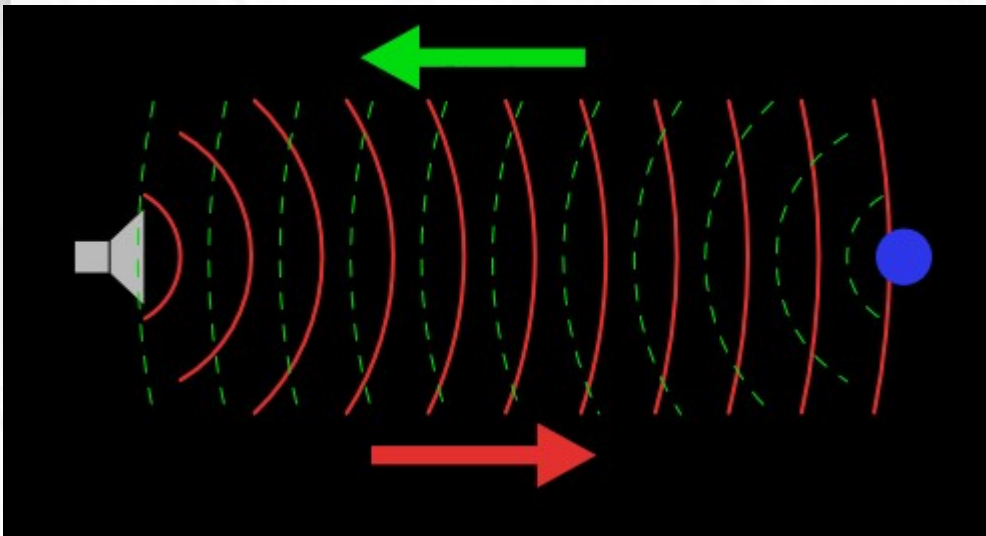


Echolocation = depth of the oceanic floor.

ECHOLOCATION

Echolocation is the process of using waves reflected from an object To determine its location.

Example: radar and sonar. Below sound waves are used to locate An object.



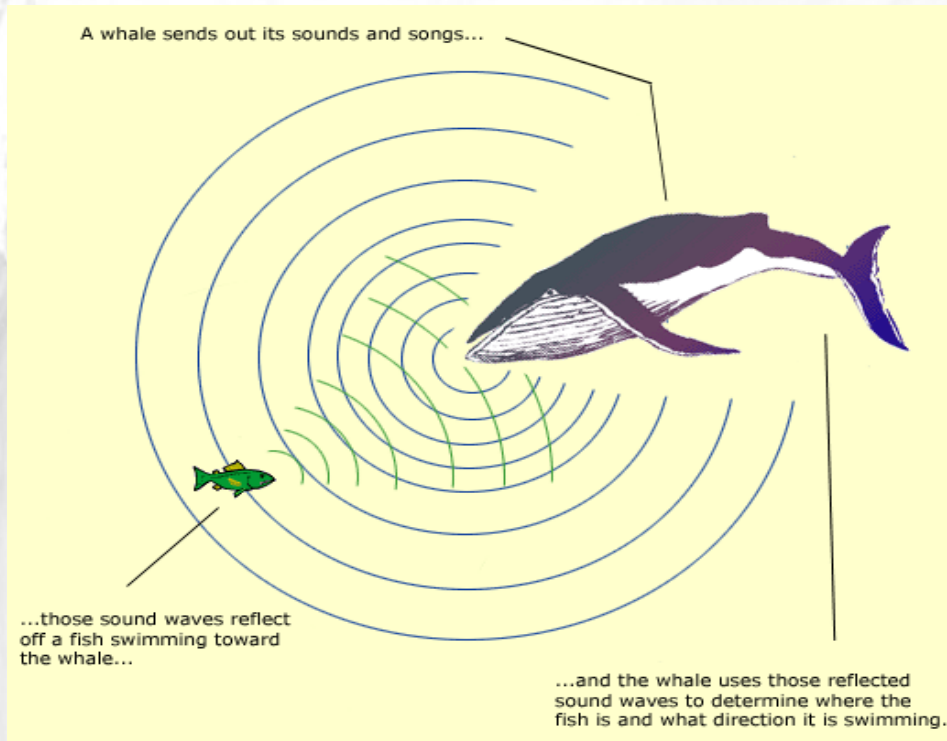
The speed of sound in Water is known.

If the time for the Wave to come back is recorded, The distance source-object is Computed. Whales , dolphin Bats use this phenomenon.

$$V= 2d/t$$

d distance source-object

SONAR echolocation



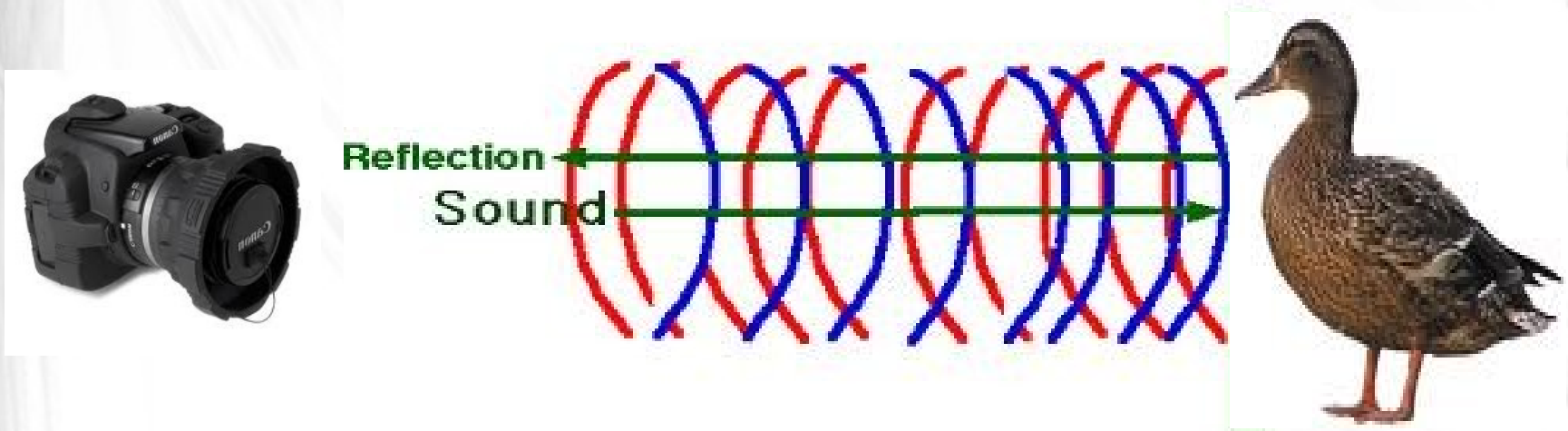
For example:
If you shout at a cliff
And hear the echo
1 second later
How far is the cliff from you.
Speed = 344m/s at room
Temperature ?

A person stands at a point 300m in front of the face of a sheer cliff.
If the person shouts, how much time will elapse before
An echo is heard ?

AUTOFOCUSING with SOUND WAVES

Auto focusing cameras emit a pulse of very high frequency (ultrasonic) sound that travels to the object, and include a sensor that detects the returning Reflected sound wave. Let's get an idea of the time sensitivity of the detector.

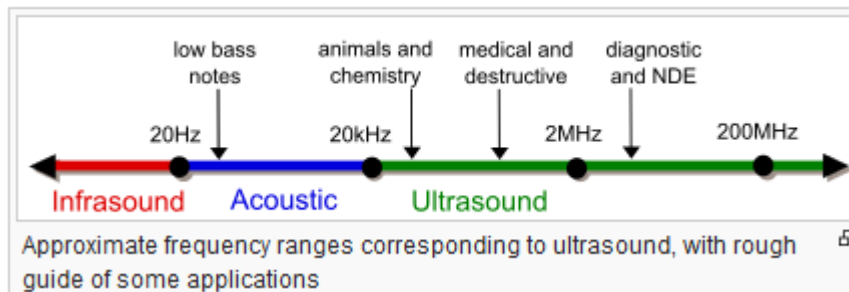
Calculate the travel time of the pulse for an object 1 m away and 20 m away.
($v=343\text{m/s}$)



The time is very short so the wait is not noticeable

1 Applications of Sound in Medicine

By scanning ultrasonic waves across the body and detecting the echoes from various locations, it is possible to obtain an image.



If a room as the shape of an ellipse. Sound produced at 1 focus Of the ellipse will reflect on the wall and will end up at the other Focus. If you whisper at 1 focus, you will be heard at the other focus.



Whispering Gallery
In Belfasts City Hall



NEW YORK CITY - *The Whispering Gallery*

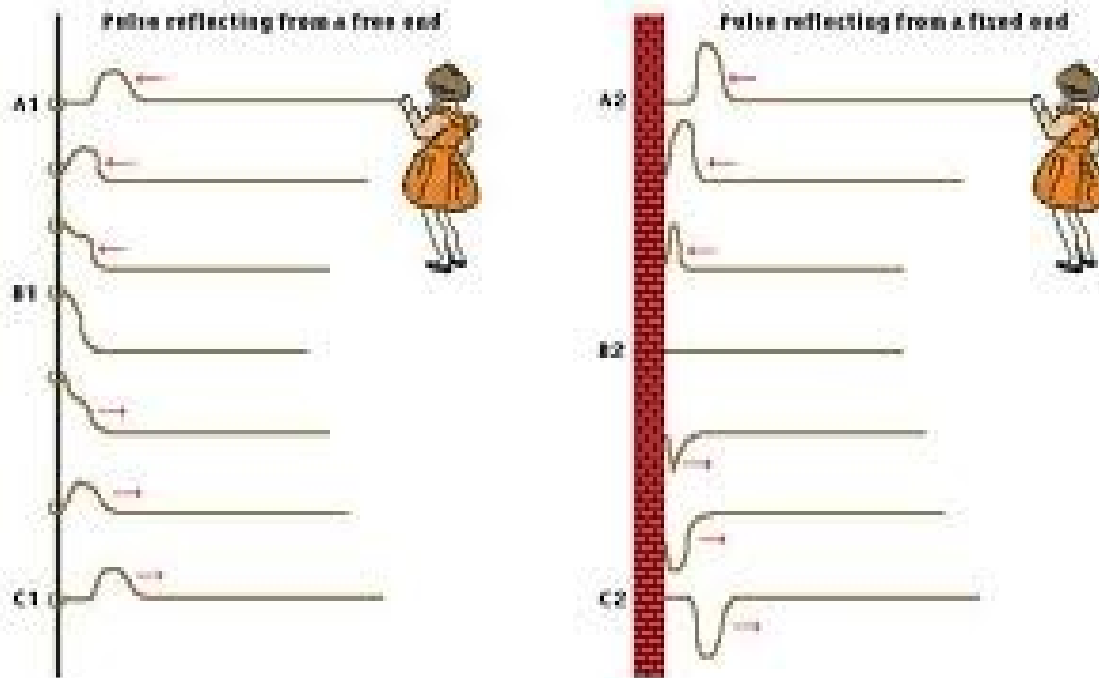
In Grand Central See Link below

<http://www.nothingtoseehere.net/things/transportation/>

I- REFLECTION

Abrupt change in the direction of propagation of a wave that strikes the boundary between different mediums.

Example: wave on a string can reflect off a boundary.



1) The huge volcanic eruption on the island of Krakatoa, Indonesia,

In 1883 was heard on Rodriguez Island, 4,782 km (2 970 miles) away. How long did it take the sound to travel to Rodriguez?

2) A sound emitted underwater reflects off a school of fish and is detected at the same place 0.01s later. How far away are the fish ? (look for speed of sound in water)

3) A sonic depth gauge is placed 5m above the ground. An ultrasound pulse sent downward reflects off snow and reaches the device 0.03 seconds after it was emitted. The air temperature is -20C.

A) How far is the surface of the snow from the device ?

B) How deep is the snow?